OPEN-SOURCE AI PROJECT

DocVerify AI Studio

Autonomous Local Document Verification & Fraud Detection Powered by Open-Weight AI

INFERENCE RUNTIME	SUPPORTED LLMS	ORCHESTRATION
100% Offline / Local	Gemma 2 • Llama 3.1 • Qwen 2.5	Multi-Agent RAG Pipeline

THE CHALLENGE

Document Fraud vs. Data Privacy Trade-off

Cloud Privacy Exposure

Sending sensitive legal, financial, or personal ID documents to third-party cloud APIs violates strict privacy regulations.

Sophisticated Forgeries

Modern digital alterations (modified amounts, dates, and names) bypass traditional static OCR systems without forensic verification.

Manual Audit Bottlenecks

Human compliance checks are slow, inconsistent, expensive, and scale poorly during high volume onboarding.

THE SOLUTION

DocVerify AI Studio Architecture

Zero-Cloud Data Sovereignty with Multi-Agent Intelligence

An open-source desktop environment where files remain entirely on local hardware while autonomous AI agents audit, extract, and verify document integrity in real time.

- **Zero Cloud Latency:** Runs on Ollama/LM Studio backends locally
- **Forensic Extraction:** Detects metadata shifts and structural tampert
- **Audit Traceability:** Real-time reasoning terminal for compliance logs

100%

Air-Gapped Offline Privacy

No external API calls, zero telemetry leakage, complete compliance with local data protection laws.

OPEN-SOURCE AI TRIAD

Core Technology Stack

1. Open-Weight Models

Gemma 2, Llama 3.1, and Qwen 2.5 optimized for precise OCR, structured JSON output, and localized reasoning.

2. Agent Framework

CrewAI & LangChain multi-agent pipelines for specialized document parsing, forensic verification, and cross-checking.

3. Local Vector RAG

ChromaDB and FAISS vector stores running in-memory for instant document searching and cited audit queries.

SEQUENTIAL PIPELINE

Multi-Agent Verification Flow

AGENT 01

Parser Agent

Extracts raw OCR text, document layout, images, and metadata structures.

AGENT 02

Forensic Agent

Analyzes EXIF metadata, font integrity, and detects visual manipulation.

AGENT 03

Cross-Checker

Validates extracted fields against business rules and flag anomalies.

OUTPUT

Audit Scorecard

Generates authenticity index, risk warnings, and cited summary reports.

KEY CAPABILITIES

Platform Features & Tooling

Model Selector & Telemetry

Live performance tracking (tokens/sec, context RAM) and instant local model switching.

Preset Document Testing

Instant evaluation presets for Passports, Bank Statements, Invoices, and Contracts.

Severity Risk Flags

Automated categorization into High Risk, Warnings, and Verified Integrity tags.

Interactive Local RAG Chat

Conversational query interface returning source page numbers and exact document citations.

IMPACT DOMAINS

Target Application Areas

Financial & Banking

KYC onboarding, loan document verification, and bank statement audit checks.

Legal & Compliance

Contract loophole identification, clause verification, and NDA risk analysis.

Public Sector & HR

Identity validation, employee background verification, and tax document checks.

OPEN SOURCE REPOSITORY AVAILABLE

Thank You!

Building a safer, privacy-first future with open-weight artificial intelligence.

github.com/rahulam230407-art/Hacktoberfest31

ollama run gemma2